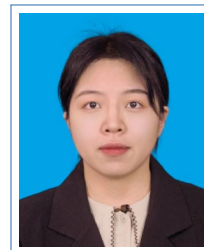


Xiyu Gu

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Education

Jan.2024 – Now

University of Padua

Ph.D. Candidate

Subject: Computing-Communication-Control Co-design

Padua, Italy

Supervisor Prof. Matthias Pezzutto

Co-supervisor Prof. Luca Schenato

June.2022 – Dec.2023.

University of Victoria

Research Assistant

Victoria, Canada

Supervisor Prof. Yang Shi

Sept.2019 – Mar.2022

Northwestern Polytechnical University

Master of Engineering

Subject: Navigation, Guidance & Control

GPA: 3.55/4.00, rank 3/69

Project 985 & 211, Xi'an, China

Supervisor Prof. Jianguo Guo

Co-supervisor Prof. Zongyi Guo

Sept.2015 – Jun.2019

Northwestern Polytechnical University

Bachelor of Engineering

Subject: Flight Vehicle Design & Engineering

GPA: 3.58/4.00, rank 2/32

Project 985 & 211, Xi'an, China

Research Interests

- Control theories, including Model Predictive Control (MPC), prescribed performance control (PPC), robust control, adaptive control, optimal control and cooperative control
- Control over wireless, including edge-based and cloud-based optimal control with packet loss and delay
- Applications, such as multi-performance requirements for flight control, cooperative control of multi-agent system, route planning

Publications

Journal Papers.....

[1] X. Gu, L. Schenato, S. Dey, M. Pezzutto, Communication-Computation Trade-off and Optimal Control Selection in LQG Control, *IEEE Control Systems Letters*, 2025.

[2] X. Gu, Z. Guo, X. Wang, et al, Prescribed Performance-Guaranteed Tracking Control: A Switched Positive System Perspective. *IEEE Control Systems Letters*, 2024.

[3] Z. Guo, X. Gu, Y. Han, et al, Performance-Guaranteed Attitude Tracking Control for RLV: A Finite-Time PPC Approach. *IEEE Transactions on Aerospace and Electronic Systems*, 2024. (**Corresponding author**)

[4] Z. Guo, D. Herry, X. Gu, et al, Maneuvering tracking algorithm for reentry vehicles with guaranteed prescribed performance, *IEEE Transactions on Aerospace and Electronic Systems*, 2024, 60.3: 3012-3020.

[5] X. Gu, J. Guo, Z. Guo, Performance improvement-oriented reentry attitude control for reusable launch vehicles with overload constraint, *ISA Transactions*, 2022, 128: 386-396.

[6] J. Guo, **X. Gu**, Z. Guo, Asymptotic adaptive tracking control for hypersonic vehicles with guaranteeing multi-performance requirements. *Aerospace Science and Technology*, 2020, 105: 106025.

Conference Papers.....

[1] **X. Gu**, S. Dey, M. Pezzutto, Cloud-based Reference Governor with Dynamic Computation Offloading over Shared Computing Resources, *European Control Conference (ECC)*, 2026.

[2] **X. Gu**, M. Pezzutto, L. Schenato, S. Dey, Optimal Control Selection over the Edge-Cloud Continuum, *IFAC- PapersOnline*, 2025. (**Finalist for Young Author Award**)

Under Review.....

[1] Y. Dong, Z. Guo, Z. Peng, **X. Gu**, Data-Physics Fusion-Guided Bayesian Fuzzy Learning for Interpretable Reentry Vehicle Trajectory Prediction, *IEEE Transactions on Aerospace and Electronic Systems*, under review, 2026. (**Corresponding author**).

[2] **X. Gu**, M. Pezzutto, L. Schenato, S. Dey, Cooperative Edge-Cloud MPC for Constrained Control under Arbitrary Communication Environment, *Automatica (regular paper)*, under review, 2025.